

M.PHARM PHARMACY ADMINISTRATION · PIMS

# Electronic Prescribing

## in Hospitals & Community

Exhaustive offline study reference · 28 sections · Full glossary · Further reading

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*Course: Pharmacy Information Management Systems (PIMS)*

|                 |                                                              |
|-----------------|--------------------------------------------------------------|
| <b>Length</b>   | ~50 pages of structured reference content                    |
| <b>Format</b>   | Designed for offline study, printing, or annotation          |
| <b>Source</b>   | Mirrors the 28-slide master lecture on GitHub Pages          |
| <b>Audience</b> | Master in Pharmacy Administration · 90-min lecture companion |
| <b>Updated</b>  | 2026 · University of Jordan                                  |

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## A · FOREWORD

# How to use this reference

This document is the **exhaustive offline companion** to the 90-minute master lecture "Electronic Prescribing in Hospitals & Community". Every slide on the live site has been expanded into prose suitable for self-paced study, exam preparation, and dissertation reference. You do not need an internet connection to use it.

The document is organised in the same nine learning blocks as the live lecture: (1) system mapping, (2) definitions and terminology, (3) core components and drivers, (4) hospital EP — benefits, evidence, and new errors, (5) Barcode Medicines Administration, (6) Clinical Decision Support, (7) workflow, usability and integration, (8) community e-prescribing and the hospital→community bridge, and (9) Jordan context, governance, and take-home messages — followed by a comprehensive glossary and further-reading section.

**How to study**

Read each section straight through once. Then return to the **Self-test prompts** in Section 25 and answer them in your own words. Where you struggle, return to the specific section. When you can answer all 18 prompts without notes, you are ready for both the assessment and the cases module.

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## B · LEARNING OUTCOMES

# What you will master

By the end of this 90-minute lecture (and this companion reference), you will be able to:

**01****Map the medicines-use cycle**

Explain where electronic prescribing fits in the prescribing → dispensing → administration → monitoring cycle and how EP differs across hospital and community settings.

|           |                                                                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>02</b> | <b>Distinguish EP, CPOE and eTP</b><br>Use precise terminology to differentiate Electronic Prescribing, Computerized Physician Order Entry and Electronic Transfer of Prescriptions — and explain why these distinctions matter clinically and contractually. |
| <b>03</b> | <b>Critically appraise the safety evidence</b><br>Interpret error-reduction figures from Bates, Fowlie, Koppel, Poon and Strom — and explain why some error rates initially rise after EP rollout.                                                            |
| <b>04</b> | <b>Apply CDS design thinking</b><br>Differentiate active vs passive Clinical Decision Support, identify graded alerts, and recognise warning fatigue as a system-design failure rather than user negligence.                                                  |
| <b>05</b> | <b>Evaluate Jordan's readiness</b><br>Apply the lecture frameworks to Jordanian hospitals (e.g., Hakeem rollout, JUH, KAUH, KHCC, RMS) and the community pharmacy sector regulated by JFDA.                                                                   |
| <b>06</b> | <b>Govern post-implementation safety</b><br>Design a post-go-live monitoring plan: error reporting, audit trails, override-rate dashboards and BCMA workaround surveillance.                                                                                  |

## SECTION 1 · SYSTEM MAPPING — THE MACROVIEW

# The medicines-use system — a macroview

Before we zoom into electronic prescribing, we must zoom *out*. EP is not a product; it is one node inside a sociotechnical network of people, paper, software, regulations, and pharmacology.

## The hospital setting — closed loop

Inside a hospital, every actor in the medicines-use cycle — prescriber, pharmacist, nurse, lab, and patient — shares **one Electronic Health Record (EHR)** and **one patient identifier** (the Medical Record Number or MRN, usually printed on the patient's wristband). This shared substrate is what makes the hospital a "closed loop":

- The prescriber logs in to the EP system through the EHR, sees the patient's allergies, lab results (eGFR, K+, INR), problem list, and current medications.
- The pharmacist verifies the order, dispenses, and supplies ward stock — all visible in the same record.
- The nurse administers the dose at the bedside, using **Barcode Medicines Administration (BCMA)** to verify the Five Rights (patient, drug, dose, route, time).
- The lab feeds results (creatinine, potassium, drug levels) back into the EHR, where they can trigger alerts in the EP module.
- Every event is timestamped and attributed in an immutable audit trail.

## The community setting — open loop

The community medicines-use system is structurally different. The prescription is portable (the patient takes it to a pharmacy of their choice), the data exchanged is much thinner, and the patient self-administers at home with no nurse present to apply BCMA.

- A General Practitioner (GP) prescribes in their practice EMR.
- An e-prescription token is issued — a QR code, barcode, or alphanumeric code on the patient's phone or paper.
- The patient walks to any community pharmacy.
- The pharmacist validates the token against a national spine (e.g., NHS EPS in the UK, SureScripts in the US, Hakeem in Jordan within public-sector facilities), dispenses, and counsels the patient.
- The patient takes the medicine home and self-administers.
- An insurance payer may verify eligibility and apply co-payment rules.

- A regulator (JFDA in Jordan) maintains controlled-drug oversight.

## Four lessons in one picture

- 1. Hospital EP is a closed loop.** Every actor shares one EHR and one patient identifier. Decision support has rich data to draw from: weight, eGFR, allergies, current INR.
- 2. Community EP is an open loop.** The prescription is portable (any pharmacy), and the data shared between GP, pharmacy, and insurer is far thinner. The CDS engine knows much less.
- 3. The bridge is the weakest link.** Discharge prescriptions must cross from the hospital EP system to the GP EMR and ultimately the community pharmacy. In most countries — and certainly in Jordan — this is still partly manual.
- 4. EP changes the visibility of risk.** Paper hides errors; electronic systems surface them. This is why error rates can appear to rise after go-live before they fall.

### Why this matters

If you only ever see EP as a screen with drug names on it, you will miss the **system**. The exam — and your future hospital director's office — will reward students who can see the whole map: people, data flows, regulators, and the gaps between them.

### Jordan context

The macroview has three real anchors in Jordan: **Hakeem** (the public-sector EHR rolled out by Electronic Health Solutions across MoH hospitals, RMS, and university hospitals); **private hospital EHRs** (Cerner, InterSystems TrakCare, and locally built systems at Istishari, Khalidi, Specialty); and **community pharmacy software** (varied, mostly dispensing-focused, rarely connected to Hakeem). The bridge between hospital discharge and community pharmacy is overwhelmingly **paper** — a printed prescription handed to the patient at discharge.

## SECTION 2 · PRE-DIGITAL PAIN POINTS

# Why we needed EP

Before electronic prescribing, the medicines-use cycle was a sequence of handovers built on paper, memory, and bleeped phone calls. Each handover was a place where errors hid. Six classic pain points explain why every modern hospital eventually replaces paper.

1

**Illegible handwriting**

The classic image of a doctor's scrawl is not a joke — it is a documented patient-safety hazard. Pharmacy callbacks to the ward to clarify "Inderal vs Inderide" or "Losec vs Lasix" were daily events. Look-alike, sound-alike (LASA) errors caused real harm. EP eliminates handwriting entirely.

2

**Incomplete prescriptions**

Missing route, missing frequency, missing duration, missing signature, missing date. Optional paper fields invite omission. Pharmacists then guess — or chase the prescriber. Structured electronic forms make all critical fields mandatory.

3

**Transcription errors**

In jurisdictions where unit clerks re-typed orders into the pharmacy system (US-style), **around 11% of all prescribing errors** were transcription errors. EP eliminates this entire category overnight by transmitting the order electronically end to end.

4

**Lost or duplicate drug charts**

Paper drug charts at the bedside got soaked, shredded, lost, or — worse — duplicated across shifts. A second chart with a slightly different regimen is a never-event waiting to happen. EP maintains a single canonical record.

5

**Slow ward-to-pharmacy communications**

A new prescription written at 9 a.m. could take 3+ hours to reach pharmacy via the porter system. First doses of antibiotics were delayed; sepsis bundles missed their golden hour. EP routes orders to pharmacy in seconds.

6

**No audit trail**

Who prescribed? Who changed the dose? Who administered? When? On paper, the answer is "ask the team" — and memory is fallible. There is no defensible record for litigation, governance, or quality improvement. EP creates a timestamped, attributable, immutable log.

**Real example**

In the UK National Reporting and Learning System (NRLS), **medication errors are the second-largest category of incidents** reported by NHS hospitals. Every prescribing error reported on paper was a missed opportunity that EP — properly designed — could have caught at the point of entry.

#### **Jordan reality check**

Walk through the medical ward of a Jordanian public hospital today and you will still see **paper drug charts at the foot of the bed** in many sites. Hakeem-enabled hospitals have moved to electronic charts, but legacy paper persists in surgical wards, ICU during downtime, and crisis surge periods. The lessons of pre-EP pain points are not history — they are this morning's reality for many of your colleagues.

## SECTION 3 · DEFINITIONS

# What is Electronic Prescribing?

**Definition (NHS Connecting for Health)**

Electronic Prescribing is *"the utilisation of electronic systems to facilitate and enhance the communication of a prescription or medicine order, aiding the choice, administration and supply of a medicine through knowledge and decision support, and providing a robust audit trail for the entire medicines use process."*

## Three capabilities baked into the definition

Read that definition carefully. It demands **three** things, not one. A system that does only one of them is not EP — it is a glorified word-processor.

- |                                             |                                                                                                                                                                                                                                                      |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>A · Communication</b>                    | Electronic transmission of the order between prescriber, pharmacy, nurse, and patient — replacing handwriting, faxing, and paper drug charts. This is the most visible feature, but on its own it is just CPOE.                                      |
| <b>B · Knowledge &amp; Decision Support</b> | Active alerts (allergy, drug-drug interactions, dose-range checks) and passive references (BNF, formulary, monographs) embedded in the prescribing workflow. This is what turns a typewriter into a clinical-intelligence partner.                   |
| <b>C · Audit Trail</b>                      | A timestamped, attributable, immutable record of every order, every override, every administration. The legal substrate for litigation defence and the data substrate for quality improvement, pharmaco-epidemiology, and antimicrobial stewardship. |

**Examiner's lens**

If a colleague says "we have electronic prescribing" but the system *only* captures orders without alerts or audit, they have **CPOE, not EP**. The definition is your weapon: ask "does it support knowledge?" and "does it audit?" If both answers are no, it is not EP.

## SECTION 4 · TERMINOLOGY

# EP vs CPOE vs eTP — clearing the confusion

These three terms are routinely treated as synonyms in casual hospital conversation. They are not. The differences shape what your system can *safely* do.

| Term                                              | Scope                                                       | Setting                    | Key features                                                                                     |
|---------------------------------------------------|-------------------------------------------------------------|----------------------------|--------------------------------------------------------------------------------------------------|
| <b>Electronic Prescribing (EP)</b>                | Full medicines cycle — prescribing, supply, administration  | Hospitals (UK term)        | Decision support, administration tracking, audit trail                                           |
| <b>Computerized Physician Order Entry (CPOE)</b>  | Data capture only — narrower                                | US hospitals (legacy term) | Order entry without integrated decision support; applies to any order (labs, imaging, drugs)     |
| <b>Electronic Transfer of Prescriptions (eTP)</b> | Transmission only — sending the script across organisations | Community / primary care   | Sends prescriptions from GP to community pharmacy electronically (e.g., NHS EPS, US SureScripts) |
| <b>e-Prescribing (generic)</b>                    | Umbrella term used loosely in policy documents              | Any setting                | Beware: the same word covers all three of the above in different contexts                        |

## Why this matters in practice

### Clinical implications

When hospital staff describe their system as "electronic prescribing", they may assume it includes decision support — but CPOE alone does not. This mismatch causes:

- Overconfidence in system safety
- Failure to implement compensating safeguards (like pharmacist screening)
- Misunderstanding of training needs

### Research implications

Studies that compare "EP systems" may actually be comparing very different functionalities. A CPOE without CDS will reduce errors very differently from full EP. Clear definitions are essential for:

- Comparable studies and meta-analyses

- Sensible vendor procurement specifications
- Realistic implementation business cases

**Jordan context**

In Jordan, vendor pitches and tender specifications often use "EP", "CPOE", and "e-prescribing" interchangeably. As a future pharmacy administrator, your first job in any procurement is to demand the vendor define *exactly* what their module does — and which of the three columns above it actually fills.

## SECTION 5 · ANATOMY OF AN EP SYSTEM

# Core components — the six pillars

An EP system has six modules. Each can succeed or fail independently. A weakness in any one of them undermines the whole system.

1

**Medicines Database**

Structured drug data with enough **granularity** to handle complex hospital regimens — not just "amoxicillin 500 mg PO TDS" but reducing-dose regimens, alternate-day schedules, dual-route (IV→PO switch), and infusion rates. The database is the foundation: every other pillar relies on the names, strengths, forms, and routes it defines.

2

**User Interface**

Screens and workflows for prescribing, reviewing, and administering. UI choices — tall-man lettering, drug images, structured fields, smart defaults, dropdown behaviour — determine error rates. A bad UI in front of good data still kills patients.

3

**Decision Support**

Allergy checks, drug-drug interaction screening, drug-disease checks, dose validation, formulary guidance. Active alerts (interruptive) and passive references (BNF, Lexicomp). A whole section of this document is devoted to CDS — see Section 15.

4

**Integration Layer**

Interfaces (often HL7 v2 or FHIR APIs) to pharmacy stock systems, EHR demographics, pathology, discharge modules, and — critically — community systems for hand-off. EP that cannot talk to the EHR is functionally crippled.

5

**Administration Module (eMAR + BCMA)**

Records who gave what, when, and how. Barcode scanning at point-of-care verifies the 5 Rights. Without this module, EP is "half a system" — orders are placed but never closed-loop verified.

6

**Reporting & Audit Trail**

Every prescribing event, edit, override, and administration is logged. Enables clinical governance, error-rate tracking, pharmaco-epidemiology research, antimicrobial stewardship dashboards, and defensible documentation for litigation.

**Architectural insight**

You will see vendors sell "EP" that is really three of these six pillars in a trench coat. Use this map as your **RFP (Request for Proposal) checklist** — does the proposed system have all six? Which are weak? What is the integration story for the missing pieces?



## SECTION 6 · DRIVERS FOR ADOPTION

# Why hospitals adopt EP — the four drivers

No CFO signs a multi-million-dinar EP contract because of pretty screens. There are four classic business cases — and understanding which one dominates in your institution shapes the proposal you write.

## Patient Safety

Medication errors cause preventable harm and avoidable readmissions. EP provides structured interventions — alerts, dose checks, audit — that paper cannot. The safety case is the most quoted, but rarely the most decisive financially because safety benefits accrue diffusely, not to any single department's budget.

## Litigation & Risk Management

Healthcare is increasingly litigious. EP creates **defensible documentation**: who prescribed, when, and on what evidence. In a malpractice claim, the audit trail is the hospital's first witness. Insurers often discount premiums for EP-enabled hospitals.

## Clinical Governance

Transparent prescribing records support audit, antimicrobial stewardship, formulary compliance, and quality improvement. Regulators (CBAHI, JCI, MoH) increasingly require these capabilities for accreditation. Without EP, a hospital cannot credibly report override rates or alert effectiveness — which JCI surveyors now expect.

## Operational Efficiency

Electronic workflows streamline pharmacy dispensing (around **–64% turnaround** in published studies), discharge prescriptions (**80 days** → **same day** in Kirby et al. UK data), and supply-chain ordering. The CFO's case lives here — efficiency gains are measurable in headcount, beds, and bed-days.

**Jordan context**

The dominant driver varies by sector. **Public hospitals** (MoH, RMS) emphasise governance and JCI/CBAHI accreditation. **Private hospitals** (Istishari, Khalidi, Specialty) emphasise operational efficiency and patient experience. **University hospitals** (JUH, KAUH) emphasise research-grade data and pharmaco-epidemiology. Knowing which driver dominates in *your* institution shapes the business case you would write.

## SECTION 7 · PROMISES AND LIMITS

# What EP promises — and what it cannot guarantee

## Core promises

- Legible, complete prescriptions — no more illegibility or missing fields
- Reduced medication errors — across multiple categories
- Decision support at the point of care — alerts and references
- Comprehensive, attributable audit trails — defensible documentation
- Streamlined ward–pharmacy workflows — faster turnaround
- Discharge summaries delivered on the day — not weeks later

## Extended benefits

- Formulary compliance and cost transparency — generic prescribing, on-formulary nudges
- Supply-chain integration — automatic re-ordering, par-level management
- Real-time KPI monitoring — antimicrobial usage, high-cost drug spend, polypharmacy rates
- Data for research, audit, and antimicrobial stewardship — pharmaco-epidemiology made possible

### Critical reality check

**EP systems are tools, not magic.** They cannot eliminate medication errors entirely. Poor design introduces new errors. Success depends on implementation, training, culture, and continuous monitoring. Specifically:

- EP ≠ error elimination — studies show 17–86% reduction depending on error type and system design
- Design determines benefit — a poorly designed system can *increase* errors
- Human factors persist — workarounds, alert fatigue, cognitive load
- Initial error rates may rise — better detection plus learning curve

### Sociotechnical lens

The single most important sentence in this lecture: **"EP is a sociotechnical system, not a software product."** Buying the software is 20% of the problem; designing the workflows, training the people, and shaping the culture is the other 80%.

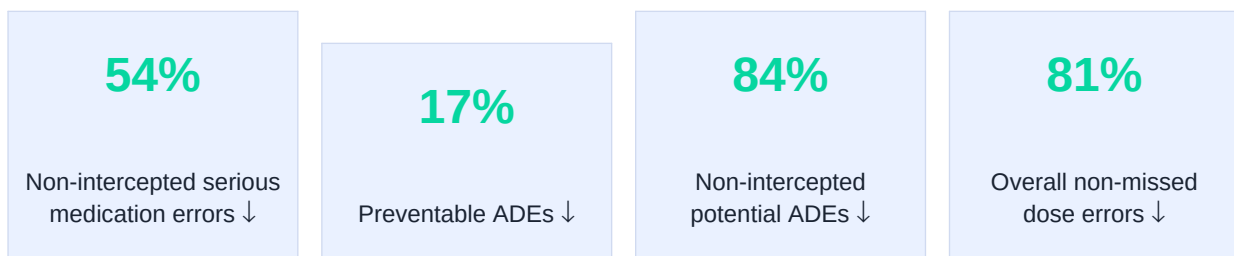
## SECTION 8 · HEADLINE EVIDENCE — HOSPITAL EP

# What does the literature actually show?

The literature is rich but heterogeneous. Three landmark studies anchor most lectures and exam questions. Know them by author, setting, headline figure, and the catch.

## 1. Bates et al. (Brigham & Women's Hospital, USA — JAMA 1998)

A landmark before-and-after study at a single large US academic hospital. The headline figures students must memorise:



### Critical reading

The 84% reduction in *potential* errors is dramatic — but a potential error is one caught before reaching the patient. The clinically meaningful figure is the **17% reduction in preventable ADEs**. Always read the denominator and the outcome carefully. Headlines mislead.

## 2. Fowle et al. (Ayrshire & Arran, UK)

A UK NHS hospital study comparing pre-EP and post-EP (12 months) error rates across three medicines-use stages.

| Setting                  | Pre-EP error rate | Post-EP (12 months) | Significance     |
|--------------------------|-------------------|---------------------|------------------|
| Inpatient prescribing    | 7.4%              | 4.7%                | p < 0.001 (sig.) |
| Discharge prescribing    | 7.5%              | 5.9%                | NS (not sig.)    |
| Medicines administration | 9.0%              | 5.4%                | p < 0.001 (sig.) |

Note that discharge prescribing already had a structured paper template, leaving little room for EP-driven improvement. **Context determines impact.** In a Jordanian context, the closest analogue is the contrast between highly structured ICU drug charts (already templated, less room

for EP gain) versus medical ward charts with dozens of free-text additions per shift (huge room for gain). Always think: where is the paper most chaotic? That is where EP delivers the biggest safety dividend.

### 3. Westbrook et al. (Sydney, Australia — PLoS Medicine 2012)

A multi-site comparison of two different CPOE systems in Australian hospitals. The key finding: **different EP systems produced different error-rate trajectories**. One system reduced procedural errors by 60% while the other reduced them by 90%. The choice of vendor, configuration, and rollout method matters as much as the decision to deploy EP at all.

#### Synthesis

"EP reduces errors" is a defensible headline. "Any EP system reduces errors equally" is wrong. Systems differ; deployments differ; outcomes differ. Always ask: *which* EP, *where*, *how* implemented, and *measured by whom*?

## SECTION 9 · THE J-CURVE

# Why error rates often rise after go-live

A near-universal pattern: errors spike for 3–6 months, then decline below baseline. Understanding why protects you from premature panic and from prematurely declaring success.

## The shape of the curve

Plot prescribing error rate on the y-axis against time-since-go-live on the x-axis. The typical trajectory:

- **Months 0–3:** errors rise sharply above the pre-EP baseline
- **Months 3–6:** errors peak, then begin to decline
- **Months 6–12:** errors fall to or below baseline
- **Months 12+:** errors stabilise at a new, lower steady state

## Three explanations for the rise

- 1. Enhanced detection** — Electronic systems capture errors that paper charts hid. The errors did not increase; the visibility did. This is mostly *good news*, even if it looks bad on the dashboard.
- 2. Learning curve** — Staff unfamiliar with new screens, dropdowns, and workflows make selection mistakes. This is real but transient — typically resolves over 3–6 months with good training.
- 3. System design flaws** — If the curve does *not* recover, you have a design problem: confusing dropdowns, look-alike defaults, missing fields. Persistent post-go-live error rises are a red flag for governance, not a learning curve.

### For administrators

Build a **30-day, 90-day, and 6-month review** into your EP rollout governance plan. Distinguish detection effects from real increases by sampling pre-EP charts retrospectively and applying the EP error definitions to them. Otherwise you are comparing apples (paper-era detection) with oranges (EP-era detection) — and the comparison is meaningless.

## SECTION 10 · IATROGENIC ERRORS

# The new errors EP creates

Koppel et al. (JAMA 2005) studied a commercial EP system at a US tertiary hospital after seven years of use. They identified **22 distinct error types** facilitated by the system. The errors fell into two families.

## Family A — Information fragmentation

Critical information is scattered across screens, modules, and windows. The prescriber cannot see the whole picture from a single view.

- Allergies on one screen, prescription on another
- Discontinued meds still listed in active orders
- Critical labs (eGFR, INR) not visible at the point of prescribing
- Free text in one field contradicting structured data in another

## Family B — Human-machine interface

The UI itself introduces selection and acceptance errors.

- Dropdowns selecting the wrong drug (look-alike names)
- Default doses accepted without adjustment
- Easier to prescribe than to discontinue → polypharmacy accumulates
- Click-fatigue overrides on important alerts

## Concrete examples you must remember

### A · hydrALazine vs hydrOXYzine

Two unrelated drugs whose names share the first six letters. In an alphabetised dropdown, they sit one above the other. The fix: **tall-man lettering** (capitalised distinguishing letters), separate them visually in dropdowns, and add a pack image where possible.

### B · Default-dose acceptance

Adult default 240 mg gentamicin pre-populates for an 18 kg paediatric patient. Junior doctor clicks "OK" without adjusting. The dose is now **10× the safe dose**. The fix: context-aware defaults (paediatric flag suppresses adult defaults), weight-based dose calculators, and high-dose interruptive alerts.

### C · Discontinuation friction

Ordering a new drug takes 4 clicks. Discontinuing an old one takes 8. Patients accumulate medications across admissions; polypharmacy creeps; deprescribing fails by design. The fix:

equalise the workflow effort — make discontinuation as cheap as prescribing.

### D · Free-text discrepancies

Magrabi et al. studied 2,914 prescriptions with free text. **16.1% had internal discrepancies** — for example, the structured field said "10 mg" but the free text said "two tablets". **83.8% of these discrepancies led to adverse events.** The fix: minimise free text. Force structured fields with validation.

#### Mental model

EP eliminates whole *categories* of error (illegibility, transcription, lost charts) and creates new ones (selection, default acceptance, fragmentation). Net safety only improves if the new errors are fewer, less frequent, and less severe than the old ones — which requires **continuous post-implementation surveillance.**

SECTION 11 · PAEDIATRIC EP

# Paediatric EP — why the stakes are higher

Paediatric prescribing is high-risk by nature: weight-based dosing, narrow therapeutic indices, age-related pharmacokinetics, and rampant off-label use. EP can help — but only if it is designed for paediatric complexity.

## Three structural reasons paediatric prescribing is harder

- **Weight-based dosing** — Doses calculated in mg/kg or mg/m². Adult-default systems are catastrophic when applied to a 15 kg child.
- **Narrow therapeutic index** — Aminoglycosides, anticonvulsants, chemotherapy — small dose deviations cause large harm.
- **Age-related PK/PD** — Neonates, infants, children, and adolescents are not "small adults". Renal and hepatic clearance vary dramatically by age.

## Evidence highlights

| Study          | Finding                                                                                   | Take-home                                                |
|----------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------|
| Ghaleb et al.  | EP reduces dosing errors and missing information in paediatrics, but adoption is slow     | Vendors under-invest in paediatric modules               |
| Jani et al.    | Error rate fell from 77.4% → 4.8% in renal paediatric patients                            | Mostly omission errors prevented by structured prompts   |
| Warwick et al. | Prescribing errors fell from 8.8% → 4.6% at 6 months; omitted doses reduced significantly | Sustained gain when system supports paediatric workflows |

### Jordan context

Jordan's tertiary paediatric care concentrates at **King Hussein Cancer Center (KHCC) paediatric oncology**, **Queen Rania Children's Hospital**, and the paediatric units at JUH and KAUH. Any EP system serving these sites must include weight-/BSA-based dosing calculators, paediatric formulary, age-banded high-dose alerts, and dilution rules for neonatal IV preparations. A generic adult-focused EP module is **unsafe** in these contexts.

### Lesson

If your hospital has a paediatric unit and your EP system does not have a dedicated paediatric module, that is a clinical risk you must register on the corporate risk register. "We bought EP" is not the same as "we made paediatric prescribing safer".

## SECTION 12 · BARCODE MEDICINES ADMINISTRATION

# BCMA — closing the loop at the bedside

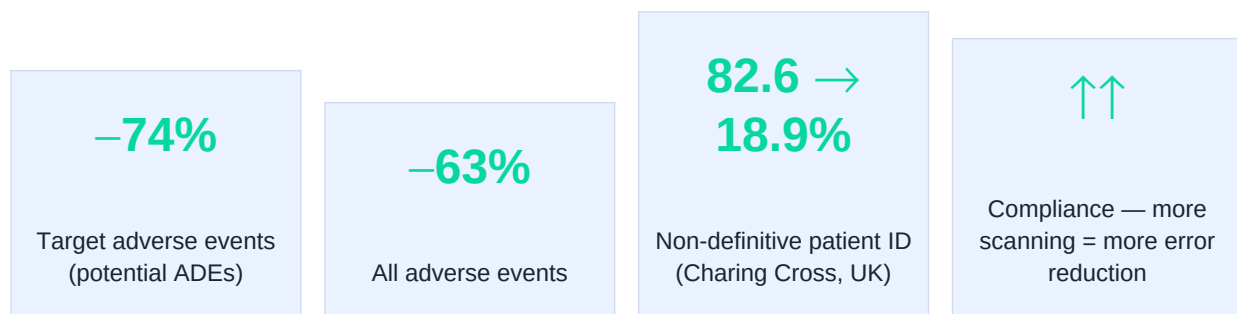
EP without BCMA is a half-loop. The closing of the loop happens at the bedside, where the nurse, the medication, and the patient meet.

## The 4-step BCMA workflow

- 1. Scan the patient.** The nurse scans the wristband barcode — the system verifies the MRN matches the order. *Right Patient ✓*
- 2. Scan the medication.** The pack barcode (GS1 / EAN / NDC) is scanned — the system verifies drug, strength, and form. *Right Drug + Dose ✓*
- 3. System verifies.** Cross-check against eMAR confirms the right route and right time. *Right Route + Time ✓*
- 4. Log the administration.** The eMAR entry is timestamped and attributed. *Audit trail ✓*

## Evidence on effectiveness — Poon et al. (NEJM 2010)

A landmark study of BCMA at a US academic medical centre, analysing 369,000 medication administrations.



### Definition refresher

The **Five Rights** of medication administration: *Right Patient · Right Drug · Right Dose · Right Route · Right Time*. Some institutions add Right Documentation and Right Reason for a "Seven Rights" framework. BCMA mechanises the first five.

## SECTION 13 · BCMA WORKAROUNDS

# Workarounds — and why they happen

Koppel et al. identified **15 distinct BCMA workarounds** across five hospitals. Workarounds are not a moral failing of nurses — they are a sign that the system, the workflow, or the culture is misaligned.

## Four root causes

- **Technical failures** — Scanners broken or unavailable. Wireless black spots. System lag. The first time the scanner fails, the nurse improvises. The tenth time, improvisation becomes the norm.
- **Time pressure** — Scanning every dose for 15 patients on a 2-hour med round adds 20 minutes. On a short-staffed night shift, that 20 minutes feels impossible.
- **Lack of awareness** — Agency staff, locums, bank nurses, new starters — anyone outside the routine training cycle may simply not know the BCMA process or its rationale.
- **Workaround culture** — The most dangerous category. If experienced nurses say "we don't really scan on this ward", the workaround becomes invisible to managers and hospital-wide.

## Common specific workarounds — and their risks

| Workaround                       | What it looks like                                                         | Risk it creates                                                              |
|----------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Batch scanning                   | Scanning all meds for all patients at the start of shift.                  | Wrong drug or wrong patient at the bedside is no longer caught.              |
| Wristband detached               | Scanning the loose wristband on the bedside table, not on the patient.     | The wrong patient receives the dose if beds were swapped.                    |
| Override "ID not confirmed"      | Bypassing the patient ID check with a verbal confirmation.                 | Defeats the entire purpose of BCMA.                                          |
| Ward-stocking pre-labelled doses | Labelling unit-dose packets ahead of time without scanning at the bedside. | Drift between physical pack and label; wrong pack reaches the wrong patient. |

## How to manage workarounds — McNulty et al.

- **Foster ownership.** Nurses must see BCMA as supporting their work, not policing it. Engage them in design, training, and post-rollout review.

- **Fix infrastructure.** Address known weaknesses: black spots, broken scanners, slow networks. Workarounds rooted in tech failures cannot be trained away.
- **Hard stops with caution.** Linking BCMA to automated dispensing cabinets so meds are inaccessible without scan can enforce compliance — but unintended consequences are real (Strom et al.).

#### Audit method

The single most powerful BCMA governance tool is the **weekly scan-rate dashboard by ward and shift**. Wards with low scan rates need a conversation, not a memo. The ward sister knows why scanning is hard — ask her, and you will find the fix.

## SECTION 14 · BCMA LIMITATIONS

# Known limits of BCMA

BCMA is powerful but not free. Its limits define the scope of complementary controls you must put in place.

|                                          |                                                                                                                                                                                                        |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hardware dependency</b>               | Scanners must be available, durable, charged, and compatible with the wireless infrastructure. A 10% scanner outage rate quickly becomes a 30% workaround rate.                                        |
| <b>Barcode availability</b>              | Not all medications carry usable barcodes. Repackaged items (compounded oncology, paediatric suspensions) require hospital-applied labels — adding cost and a new error pathway at the labelling step. |
| <b>Standards harmonisation</b>           | Different barcode standards (EAN-13, GS1 DataMatrix, NDC) coexist. Interoperability issues affect smaller hospitals dependent on multiple suppliers.                                                   |
| <b>Technology evolution</b>              | RFID, NFC, and computer-vision (camera-based pack recognition) may eventually replace 1D barcodes. Investment cycles must anticipate this.                                                             |
| <b>Original-pack dispensing required</b> | BCMA works best when manufacturer packaging reaches the bedside. This conflicts with traditional ward pharmacy practices that decant into smaller containers.                                          |
| <b>Liquid and IV preparations</b>        | Compounded IVs, paediatric suspensions, and patient-specific TPN bags need pharmacy-printed barcodes that survive the supply chain.                                                                    |

**Jordan reality**

For Jordanian hospitals planning BCMA, the practical first audit is: **what fraction of doses leaving your central pharmacy carry a usable, scannable barcode?** If the answer is below 70%, BCMA cannot be your primary safety net — it has to be paired with double-checks and pharmacist ward review.

## SECTION 15 · CLINICAL DECISION SUPPORT

## CDS — active vs passive

Clinical Decision Support (CDS) is what turns EP from a typewriter into a clinical-intelligence partner. There are two flavours, both essential, neither sufficient alone.

| Dimension          | Active CDS                                                                            | Passive CDS                                                                     |
|--------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Definition</b>  | Automatic alerts during workflow without the user seeking information                 | Reference sources accessible via links when the user actively seeks information |
| <b>Examples</b>    | Allergy alerts, drug-drug interactions, dose-range checks, duplicate-therapy warnings | BNF online, UpToDate, Lexicomp, local formulary, hot-key links                  |
| <b>Integration</b> | Built into the EP software — interruptive                                             | Stand-alone resources on intranet or external sites                             |
| <b>Strength</b>    | Catches errors the prescriber might miss; enforces baseline safety                    | Provides depth for complex decisions without interruption                       |
| <b>Weakness</b>    | Risk of <b>alert fatigue</b> if poorly configured                                     | Relies on the user recognising when they need information; easily ignored       |

### What good active CDS does well

- **Allergy checking** — Cross-reference patient allergies against prescribed medications; flag sensitivities before harm occurs.
- **Drug-drug interactions** — Identify pharmacokinetic and pharmacodynamic interactions; prioritise by severity (graded interactions).
- **Duplicate therapy** — Detect two drugs from the same class (e.g., two NSAIDs) or the same drug under different brand names.
- **Dose validation** — Check doses against maximum recommended limits; adjust for renal or hepatic function; flag high-risk dosing.
- **Formulary guidance** — Steer prescribers toward formulary-approved options; display cost; promote generic prescribing.
- **Monitoring prompts** — Remind clinicians to order necessary tests (INR for warfarin, digoxin levels, K<sup>+</sup> for ACE inhibitors).

**Design heuristic**

Active CDS *catches*; passive CDS *deepens*. Both are needed. A system that only does active alerts becomes a noisy nag; a system that only does passive references becomes a dictionary the busy doctor never opens.

## SECTION 16 · WARNING FATIGUE

# Alert fatigue — the silent killer of CDS

**Quotable figure (Huntemann et al.)**

**97% of allergy alerts were overridden** on a real EP system in routine operation. If clinicians dismiss 97 out of 100 alerts, the system has lost credibility. The 3 alerts that mattered are buried under 97 that did not. The next time a real alert fires, it goes the same way as the rest: *click, override, continue*.

## Causes of warning fatigue

- **Low specificity** — Alerts fire for clinically insignificant interactions (e.g., antacid reducing absorption of azithromycin) at the same level as catastrophic ones (clarithromycin + simvastatin → rhabdomyolysis).
- **Poor prioritisation** — Minor warnings presented with the same red font, the same modal pop-up, the same override-justification field as severe ones. The brain learns to ignore.
- **Interruptive design** — Every alert blocks the workflow and demands an action. With 30+ alerts per shift, the prescriber's only survival strategy is reflexive override.
- **No specialty tuning** — An oncology resident receives the same alerts as a paediatrician, even though their drug exposures are radically different. Untuned rules destroy specificity.

## The fix — graded alerts

Reserve the strongest interruption for the highest harm. Most alerts should never interrupt the prescriber. A three-tier model works well:

| Tier                        | When it fires                                                        | Required action                        |
|-----------------------------|----------------------------------------------------------------------|----------------------------------------|
| ■ <b>HARD STOP / SEVERE</b> | Catastrophic harm if ignored (e.g., IT vincristine, daily oral MTX)  | Cannot proceed without senior approval |
| ■ <b>INTERRUPTIVE</b>       | Significant risk needing review (e.g., clarithromycin + simvastatin) | Justify override in free text          |
| ■ <b>PASSIVE / FYI</b>      | Useful information, no harm risk (e.g., antacid + azithromycin)      | Silently logged; visible on review     |

**Governance KPI**

Track **override rates per alert type**. Any alert overridden >80% of the time is not contributing to safety — it is contributing to fatigue. Either tune the rule, lower its severity tier, or retire it.

## SECTION 17 · HARD STOPS

# Hard stops vs clinical judgment

A **hard stop** prevents a prescription from being entered at all — the system disallows the order. Should any prescription be absolutely forbidden? The literature gives a nuanced answer.

## The case for hard stops

Some prescriptions are unequivocally dangerous — "never events" that should never happen.

- Intrathecal vincristine (always lethal — never given IT)
- Daily oral methotrexate for rheumatoid arthritis (should be weekly)
- Adult-dose paracetamol for a neonate
- QEHB Birmingham disallows orders for documented serious sensitivities

When the harm is catastrophic and irreversible, the system should refuse.

## The case against hard stops

Strom et al. studied a hard stop for warfarin + co-trimoxazole. The alert was effective — but caused clinically significant treatment delays.

- Delays in necessary treatment
- Frustrate clinicians who know the context
- Drive workarounds (handwritten scripts, defeating EP)
- The Strom trial was **terminated early** due to harm from delay

Reserve hard stops for absolute contraindications. Use softer alerts (override-with-reason) for relative ones.

### Decision rule

Use a hard stop only when **both** of these are true: (1) the harm is catastrophic and irreversible, AND (2) there is no plausible legitimate clinical scenario in which this prescription is correct. If either is uncertain, use an interruptive alert with override-justification — not a hard stop.

## The 2×2 trade-off matrix

Visualise CDS design choices as a 2×2 of safety and usability. The target is the upper-right quadrant — **high safety AND high usability**.

|             | Low usability                                   | High usability                           |
|-------------|-------------------------------------------------|------------------------------------------|
| High safety | Hard stops, modal alerts — safe but obstructive | Smart graded alerts — THE TARGET         |
| Low safety  | Worst case — slow AND unsafe                    | Fast but unsafe — the dangerous quadrant |

## SECTION 18 · WORKFLOW DESIGN

# Why workflow matters more than the screen

The best EP system with comprehensive CDS will fail if the workflow is clunky, unintuitive, or misaligned with clinical reality. Workflow determines whether clinicians embrace or subvert the system.

| Risk                     | Cause                                                                    | Solution                                                 |
|--------------------------|--------------------------------------------------------------------------|----------------------------------------------------------|
| Incomplete prescriptions | Optional fields allow prescribers to skip critical details               | Forced prompts; require all fields before signing        |
| Incorrect drug selection | Look-alike names in dropdowns; ambiguous labels                          | Tall-man lettering; separate similar drugs; pack images  |
| Excessive steps          | 12-stage prescribing process burdens busy clinicians                     | Pre-Defined Orders (PDOs) for common prescriptions       |
| Workarounds              | System cannot handle complex regimens (alternate routes, reducing doses) | Ensure database granularity supports hospital complexity |
| Free-text discrepancies  | Free text conflicts with structured data                                 | Minimise free text; structured fields with validation    |
| Long login times         | Stringent security forces re-auth every 5 minutes                        | Smartcard tap-and-go; sensible session timeouts          |

## Administration workflow — four design choices that matter

- **Time windows.** System defines when an order is "due" (e.g.,  $\pm 1$  hour of scheduled time) and when administration can be recorded. Too narrow → late doses; too wide → off-schedule drift.
- **Missed-dose codes.** Structured reasons for non-administration (patient refused, drug unavailable, NPO for procedure, clinical decision). Free-text "not given" is useless to audit.
- **PRN lockouts.** Prevent paracetamol administration more frequently than q4h or >4 g/24h. Safety embedded into workflow, not relying on nurse vigilance alone.
- **On-hold / off-hold.** Suspend prescriptions pending results (hold warfarin until INR available). Useful for elective treatments — chemotherapy, anticoagulants, electrolyte correction.

### Design principle

Every field that is optional will be skipped by someone, in some shift. Every workflow that takes more than 90 seconds for a routine task will breed workarounds. **Design for the worst day, not the demo day.**

SECTION 19 · INTEGRATION

# EP and the EHR — integration as a multiplier

EP systems work best when integrated with the hospital's electronic health record (EHR). Stand-alone EP is functionally inferior; integration unlocks the system's intelligence.

## What integration enables

- **Comprehensive prescribing history.** Prior admissions, outpatient prescriptions, GP records — all visible at the point of new prescribing. Reduces duplication and drug-disease conflicts.
- **CDS using patient data.** Allergy alerts use the EHR allergy list. Renal-dosing alerts use the latest eGFR. Drug-disease checks use the problem list. Disconnected EP cannot do any of this safely.
- **Automatic monitoring orders.** Prescribing warfarin auto-orders an INR. Prescribing vancomycin auto-prompts for trough levels. Closed-loop monitoring reduces preventable harm.
- **Posting of lab results.** Lab results post back into EP and trigger alerts: "K<sup>+</sup> now 5.8 mmol/L — review ramipril". Real-time clinical surveillance.

## The interoperability standards stack

|             |                                                                              |
|-------------|------------------------------------------------------------------------------|
| HL7 v2 / v3 | Health Level 7 messaging — the legacy backbone for hospital integrations     |
| FHIR        | Fast Healthcare Interoperability Resources — modern API standard, JSON-based |
| SNOMED CT   | Clinical terminology — coded concepts for diagnoses, procedures, allergies   |
| dm+d (UK)   | Dictionary of Medicines and Devices — structured drug catalogue              |
| RxNorm (US) | Standardised drug naming for clinical systems                                |
| LOINC       | Logical Observation Identifiers — lab and clinical observations              |
| NDC (US)    | National Drug Code — pack-level identification                               |
| GS1 / EAN   | Global barcode standards used for BCMA                                       |

Jordan context

**Hakeem** is built on InterSystems TrakCare with a centralised database. Within Hakeem-enabled facilities, EHR–EP integration is mature: allergies, problems, and labs feed CDS in real time. Across non-Hakeem hospitals (most private chains), each facility has its own integration story; HL7 v2 is the dominant pattern, FHIR APIs are emerging. Cross-facility integration via the **National Health Information Exchange** is the next horizon.

#### Procurement question

When evaluating EP vendors, the most important integration question is not "do you support HL7?" — every vendor will say yes. Ask: **"Show me a live FHIR endpoint exchanging allergies, labs, and meds with our existing EHR in our test environment."** Demand evidence, not promises.

## SECTION 20 · COMMUNITY EP

# Extending EP into the community — a different beast

Hospital EP is a closed loop. Community e-prescribing is an open loop. The patient leaves the clinic with an electronic token and chooses any pharmacy. The architecture, the risks, and the regulation all change.

| Dimension          | Hospital EP                          | Community e-Prescribing (eTP)                     |
|--------------------|--------------------------------------|---------------------------------------------------|
| Loop               | Closed — same institution, same EHR  | Open — GP and pharmacy are separate organisations |
| Patient identifier | Hospital MRN on a wristband          | National ID, insurance number, or token           |
| Data sharing       | Rich (allergies, labs, problem list) | Thin (drug name, dose, duration only)             |
| CDS depth          | Patient-data-driven                  | Mostly drug-drug rules; limited patient context   |
| Administration     | Nurse with BCMA at bedside           | Self-administration at home                       |
| Safety net         | Pharmacist + nurse + physician       | Community pharmacist alone — often briefly        |
| Regulators         | Hospital governance + accreditation  | JFDA, MoH, insurance, controlled-drug regs        |
| Repeat prescribing | N/A (acute episode)                  | Core feature — chronic-disease management         |

## The community e-prescribing flow

The "token" travels with the patient; the data travels via a national spine. Both must arrive at the same pharmacy.

- GP / Clinician** issues an e-prescription in their Practice EMR.
- A digital **e-prescription token** is generated — QR code, alphanumeric, or barcode — on the patient's phone or paper.
- The **patient** chooses any community pharmacy.
- The **community pharmacy** validates the token against a National PMP / eTP Spine, dispenses, counsels, and logs the supply.

### Architecture insight

In community EP, the patient is the messenger of the token, but the data must arrive electronically through a trusted central spine. Without the spine, the system becomes a fancy printed prescription — losing CDS, audit, and pharmacovigilance benefits.

## SECTION 21 · eTP MODELS WORLDWIDE

# Four archetypes of community e-prescribing

Understanding the dominant world models helps you place Jordan and design future-state services.

## NHS EPS (Electronic Prescription Service) — UK

GPs send prescriptions to a national spine. Patients nominate one pharmacy ahead of time, or the prescription token goes with them. Highly mature; most repeat prescriptions are now paperless. Strengths: universal coverage, deep integration with the GP system. Limits: still patchy support for controlled drugs and some specialist regimens.

## SureScripts — USA

A private intermediary network connects EHRs to pharmacy chains. Volumes are enormous. Coverage of controlled substances (EPCS — Electronic Prescribing of Controlled Substances) is now standard in most states. Strengths: market reach, vendor neutrality. Limits: fragmented by state laws and insurance plans.

## Estonia / Finland / Denmark — National e-Prescription

Government-run, single-national-spine models with universal coverage. The patient picks up at any pharmacy with national ID. Cross-border pilots underway in the EU. Strengths: simplicity, equity, strong audit. Limits: requires national identity infrastructure.

## Hybrid / Emerging — Jordan, GCC

The public sector has Hakeem within facilities, but cross-organisational e-prescription to community pharmacies is largely paper. Insurance pre-authorisation and CHIP / claims systems are partly digital. Strengths: strong hospital EP foundation. Limits: missing national e-prescribing spine.

## What every mature eTP model requires

1. **A single, trusted patient identifier** across organisations (national ID, insurance number, or NHS-style number).
2. **A central spine or trusted exchange** with high availability and strong audit logging.
3. **Authentication standards** — digital signature for the prescriber, professional registration for the pharmacist.

**4. Controlled-drug protocols** — special handling for narcotics, psychotropics, codeine combinations. Often a separate workflow with stronger authentication.

**5. A real-time medication record** that both prescribers and dispensers can read (a "shared record" or PMR).

**6. Patient-facing access** — a portal or app where patients can see their own prescriptions and choose pharmacies.

#### **For Jordan**

The technical pieces exist (Hakeem inside hospitals, the National ID, JFDA registration of pharmacies, insurance claims platforms). What is missing is the **connecting tissue**: a national e-prescribing standard, a controlled-drug e-prescription workflow, and a patient-facing portal across sectors. This is where your generation of pharmacy administrators will lead.

## SECTION 22 · HOSPITAL → COMMUNITY

# The discharge transition — where systems most often fail

A patient leaving hospital is at peak medication risk: new drugs, stopped drugs, dose changes, and a handover to a primary-care team that may have no visibility of any of it.

## What can go wrong

- **Discharge summary delays.** Even when generated electronically, summaries can sit in batch queues for days before reaching the GP. Kirby et al. UK data: **80 days under paper, <1 day with proper EP integration.**
- **Medication reconciliation failures.** Pre-admission medications inadvertently restarted; new drugs not communicated; dose changes lost in transit. National Patient Safety Agency data: medication errors at the hospital-community interface are one of the largest causes of preventable readmission.
- **Patient empty-handed.** Patient discharged Friday afternoon with prescription that GP cannot fulfil over weekend. Patient runs out of medicines and presents to A&E on Monday.
- **Granularity loss.** Hospital EP can capture reducing-dose regimens, alternate routes, IV-to-PO switches. Community systems often cannot receive this granular data — information is silently flattened in transit.

## What a good discharge transition looks like

1. **Med rec on admission** — Pharmacist verifies pre-admission meds against patient or GP record.
2. **Inpatient EP** — All changes captured in EP with rationale.
3. **Discharge med rec** — Pharmacist reviews and reconciles before discharge.
4. **Real-time electronic discharge summary** — Transmitted same-day to GP.
5. **Patient counselling** — Pharmacist counsels patient with a simplified med list.
6. **Community follow-up** — Community pharmacist receives data, supports adherence.

### Jordan reality

In Jordan today, the most reliable hospital-to-community handover is via the **discharge prescription printed on hospital letterhead and physically given to the patient**. The patient takes it to a community pharmacy of their choice. This works for simple cases. It fails for: complex tapering regimens, controlled drugs, oncology supportive care, and any patient with cognitive impairment. The opportunity cost of not having electronic discharge to community pharmacies is measurable in readmissions.

**For your dissertation**

The hospital-community medication transition is one of the highest-yield research topics for any Master in Pharmacy Administration thesis in Jordan today. There is published international literature; there is almost no Jordanian-context data. *This is a gap waiting for you.*

## SECTION 23 · JORDAN LANDSCAPE

# Jordan's electronic prescribing landscape

A practical map for the working pharmacy administrator in Amman, Irbid, or Aqaba.

## Public sector — the Hakeem ecosystem

|                                          |                                                                                                                                                                                                                                                   |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Electronic Health Solutions (EHS)</b> | The national operator of Hakeem, the country's flagship EHR with EP and dispensing modules. Built on InterSystems TrakCare. Rollout includes MoH hospitals, RMS military hospitals, university hospitals, and primary-care comprehensive centres. |
| <b>University Hospitals</b>              | Jordan University Hospital (JUH) and King Abdullah University Hospital (KAUH) have integrated Hakeem-based EP with their teaching, research, and pharmaco-epidemiology workflows.                                                                 |
| <b>Royal Medical Services (RMS)</b>      | Military health system covering active service personnel and their families. King Hussein Medical Center is the flagship; EP rollout closely tracks Hakeem.                                                                                       |
| <b>Ministry of Health (MoH)</b>          | Primary-care centres, district hospitals, specialised facilities. Adoption uneven; rural sites still partly paper. Antimicrobial stewardship and chronic-disease registries are key data outputs.                                                 |

## Private sector — heterogeneous and competitive

|                                          |                                                                                                                                                                                                                             |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Major private hospitals</b>           | Istishari, Khalidi, Specialty, Jordan Hospital, Arab Medical Center — each runs its own EHR/EP (Cerner, TrakCare, locally built systems). Patient-facing portals and apps emerging. Limited cross-facility data exchange.   |
| <b>King Hussein Cancer Center (KHCC)</b> | JCI-accredited oncology centre with bespoke chemotherapy EP, BCMA, and pharmacist-driven medication review. The Jordanian benchmark for closed-loop hospital EP.                                                            |
| <b>Community pharmacies</b>              | Approximately 3,000+ private community pharmacies. Dispensing software is common but rarely connected to a national prescribing spine. Most prescriptions still arrive on paper.                                            |
| <b>Insurance &amp; payers</b>            | Public (Civil Insurance, Royal Court, RMS), private (MedNet, GIG, NextCare, Globemed), and self-pay. Each with its own claims and pre-authorisation workflows. EP integration with payers is partial and varies by insurer. |

## Regulators and special programmes

- **Jordan Food & Drug Administration (JFDA)** — drug registration, controlled-drug regulation, post-marketing surveillance, pharmacovigilance reporting.
- **National Center for Diabetes, Endocrinology and Genetics (NCD)** — chronic-disease registries; potential anchor for population-level e-prescribing of insulin and oral hypoglycaemics.
- **Jordan Pharmacists Association (JPhA)** — professional body; advocates for community-pharmacy integration with national e-prescribing.
- **Higher Health Council** — strategic oversight of Jordan's national e-health strategy.

#### Where we stand

Hospital EP is mature within Hakeem-enabled facilities and at KHCC. Community e-prescribing is at an earlier stage — paper still dominates. Insurance pre-authorisation is partly digital. The next strategic horizon is a **national e-prescription spine** connecting GP, hospital, community pharmacy, and payer — with strong patient-facing access and controlled-drug protocols. The skills you build in this course (definitions, evidence appraisal, CDS design, governance) are exactly the skills that will design that horizon.

## SECTION 24 · TAKE-HOME MESSAGES

# Five sociotechnical lessons

If you remember nothing else from these 90 minutes, remember these five sentences.

**01****EP ≠ plug-and-play**

EP systems require continuous monitoring, refinement, and adaptation to local context. Technology alone does not solve prescribing problems.

**02****Design determines benefit**

Poor design introduces new errors. User interface, workflow logic, and decision-support quality are the decisive factors in safety outcomes.

**03****Culture matters as much as code**

User engagement, training, and ownership determine whether EP systems are embraced or subverted through workarounds.

**04****Integrate, don't isolate**

EP works best when integrated with the EHR, the pharmacy system, and the discharge process — never as a standalone tool.

**05****Monitor and adapt**

Post-implementation surveillance is essential. Identify unintended consequences, track error patterns, and iterate system design.

**The pharmacist is the integrator**

Across hospital and community, the pharmacist is the only professional with both clinical knowledge and a daily stake in the medicines-data loop. EP succeeds when pharmacy leadership shapes it.

*EP systems have immense potential to improve patient safety, streamline workflows, and enhance care quality. Realising that potential requires more than software — it demands thoughtful design, robust implementation, and ongoing commitment to improvement.*

## SECTION 25 · SELF-TEST PROMPTS

# Self-test — 18 exam-grade prompts

Answer in your own words, citing the section number and named authors where relevant. When you can answer all 18 without notes, you are ready.

- |            |                                                                                                                                          |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q01</b> | Define EP precisely. Why is it different from CPOE? Give the three capabilities baked into the NHS Connecting for Health definition.     |
| <b>Q02</b> | Draw the macroview of the medicines-use system. Mark where the closed loop (hospital) meets the open loop (community). Label the bridge. |
| <b>Q03</b> | List the six pillars of an EP system. Which two are most commonly missing in vendor-sold "EP" products?                                  |
| <b>Q04</b> | Name the four classic drivers for EP adoption. Which dominates in public Jordanian hospitals? Which in private?                          |
| <b>Q05</b> | Quote the four headline figures from Bates et al. (1998). Which is the most clinically meaningful, and why?                              |
| <b>Q06</b> | Draw and explain the J-curve. Give three reasons error rates rise after go-live.                                                         |
| <b>Q07</b> | Describe two error types Koppel et al. identified. For each, propose a design fix.                                                       |
| <b>Q08</b> | Why is paediatric EP harder than adult EP? Give three structural reasons. Name two Jordanian tertiary paediatric centres.                |
| <b>Q09</b> | Describe the four steps of a BCMA workflow. What does each step verify in the Five Rights?                                               |
| <b>Q10</b> | Quote two evidence figures from Poon et al. (2010). What was the comparator?                                                             |
| <b>Q11</b> | Name three BCMA workarounds. For each, name the underlying root cause.                                                                   |
| <b>Q12</b> | Differentiate active and passive CDS with examples. Why are both needed?                                                                 |
| <b>Q13</b> | Why does alert fatigue happen? Describe the graded-alerts solution. Quote a published override-rate figure.                              |
| <b>Q14</b> | Argue for and against hard stops. State the decision rule for when a hard stop is justified.                                             |
| <b>Q15</b> | List the six interoperability standards every pharmacy administrator should know. Briefly state what each does.                          |
| <b>Q16</b> | Compare hospital EP and community e-prescribing across eight dimensions.                                                                 |

- 
- Q17** Describe the four global archetypes of eTP. Where does Jordan currently sit, and what must be built next?
- 
- Q18** Outline a 6-step "good" hospital→community discharge transition. What is the published paper-vs-EP discharge-summary delay (Kirby et al.)?
-

## SECTION 26 · ABBREVIATIONS GLOSSARY

# Every acronym used in this lecture

In alphabetical order. Bookmark this page.

|                   |                                                                  |               |                                                                 |
|-------------------|------------------------------------------------------------------|---------------|-----------------------------------------------------------------|
| <b>ADE</b>        | Adverse Drug Event                                               | <b>ICU</b>    | Intensive Care Unit                                             |
| <b>ADR</b>        | Adverse Drug Reaction                                            | <b>INR</b>    | International Normalised Ratio (warfarin monitoring)            |
| <b>AMS</b>        | Antimicrobial Stewardship                                        | <b>IT</b>     | Intrathecal (administration route)                              |
| <b>API</b>        | Application Programming Interface                                | <b>JCI</b>    | Joint Commission International                                  |
| <b>BCMA</b>       | Barcode Medicines Administration                                 | <b>JFDA</b>   | Jordan Food & Drug Administration                               |
| <b>BNF</b>        | British National Formulary                                       | <b>JPhA</b>   | Jordan Pharmacists Association                                  |
| <b>BSA</b>        | Body Surface Area (used for paediatric/oncology dosing)          | <b>JUH</b>    | Jordan University Hospital                                      |
| <b>CBAHI</b>      | Saudi Central Board for Accreditation of Healthcare Institutions | <b>KAUH</b>   | King Abdullah University Hospital                               |
| <b>CDA</b>        | Clinical Document Architecture (HL7 standard)                    | <b>KHCC</b>   | King Hussein Cancer Center                                      |
| <b>CDS / CDSS</b> | Clinical Decision Support / System                               | <b>LOINC</b>  | Logical Observation Identifiers Names and Codes                 |
| <b>CPOE</b>       | Computerized Physician Order Entry                               | <b>MoH</b>    | Ministry of Health                                              |
| <b>DDI</b>        | Drug-Drug Interaction                                            | <b>MRN</b>    | Medical Record Number                                           |
| <b>dm+d</b>       | Dictionary of Medicines and Devices (UK)                         | <b>MTX</b>    | Methotrexate                                                    |
| <b>EAN</b>        | European Article Number (legacy barcode)                         | <b>NCD</b>    | National Center for Diabetes, Endocrinology & Genetics (Jordan) |
| <b>eDS</b>        | Electronic Discharge Summary                                     | <b>NDC</b>    | National Drug Code (US)                                         |
| <b>eGFR</b>       | Estimated Glomerular Filtration Rate                             | <b>NHS</b>    | National Health Service (UK)                                    |
| <b>EHR / EMR</b>  | Electronic Health Record / Electronic Medical Record             | <b>NPSA</b>   | National Patient Safety Agency (UK)                             |
| <b>EHS</b>        | Electronic Health Solutions (operator of Hakeem, Jordan)         | <b>NSAID</b>  | Non-Steroidal Anti-Inflammatory Drug                            |
| <b>eMAR</b>       | Electronic Medication Administration Record                      | <b>PDO</b>    | Pre-Defined Order                                               |
| <b>EP</b>         | Electronic Prescribing                                           | <b>PMP</b>    | Prescription Monitoring Programme / Patient Medication Profile  |
| <b>EPCS</b>       | Electronic Prescribing of Controlled Substances (US)             | <b>PRN</b>    | Pro re nata — "as required" dosing                              |
| <b>EPS</b>        | Electronic Prescription Service (NHS)                            | <b>QEHB</b>   | Queen Elizabeth Hospital Birmingham                             |
| <b>eTP</b>        | Electronic Transfer of Prescriptions                             | <b>RFID</b>   | Radio-Frequency Identification                                  |
| <b>FHIR</b>       | Fast Healthcare Interoperability Resources                       | <b>RMS</b>    | Royal Medical Services (Jordan military health)                 |
| <b>GP</b>         | General Practitioner                                             | <b>RxNorm</b> | Standardised drug nomenclature (US)                             |

|            |                                                       |                  |                                                        |
|------------|-------------------------------------------------------|------------------|--------------------------------------------------------|
| <b>GS1</b> | Global Standards 1 — barcode and identifier standards | <b>SNOMED CT</b> | Systematized Nomenclature of Medicine — Clinical Terms |
| <b>HL7</b> | Health Level 7 — clinical messaging standard          | <b>TPN</b>       | Total Parenteral Nutrition                             |

## SECTION 27 · FURTHER READING

# Landmark papers

These are the papers your assessment will reward you for citing accurately. Read at least the abstract and methods of each; bookmark them in your research library.

- Bates DW, Leape LL, Cullen DJ, et al. Effect of computerized physician order entry and a team intervention on prevention of serious medication errors. *JAMA* 1998;280:1311–6.
- Koppel R, Metlay JP, Cohen A, et al. Role of computerized physician order entry systems in facilitating medication errors. *JAMA* 2005;293:1197–203.
- Poon EG, Keohane CA, Yoon CS, et al. Effect of bar-code technology on the safety of medication administration. *NEJM* 2010;362:1698–707.
- Strom BL, Schinnar R, Abera F, et al. Unintended effects of a computerized physician order entry near hard-stop alert. *Arch Intern Med* 2010;170:1578–83.
- Magrabi F, Ammenwerth E, Hyppönen H, et al. Why is it so difficult to measure the impact of clinical information systems? *JAMIA* 2011;18:S118–22.
- Fowle F, Bennie M, Jardine G, et al. Evaluation of an electronic prescribing system in a UK hospital. *BMJ Qual Saf* series.
- Westbrook JL, Reckmann M, Li L, et al. Effects of two CPOE systems on prescribing error rates. *PLoS Med* 2012;9:e1001164.
- Ghaleb MA, Barber N, Franklin BD, Wong ICK. The incidence and nature of prescribing and medication administration errors in paediatric inpatients. *Arch Dis Child* 2010;95:113–8.
- Jani YH, Ghaleb MA, Marks SD, Cope J, Barber N, Wong ICK. Electronic prescribing reduced prescribing errors in a pediatric renal outpatient clinic. *J Pediatr* 2008;152:214–8.
- Warwick C, Cooke J, Cattini P, et al. Implementation of electronic prescribing in children's services. (Multiple paediatric EP studies — see UK NHS Digital paediatric EP archive.)
- McNulty J, Donnelly E, Iorio K. Methods used by nurses to verify wristbands. *J Hosp Med* — BCMA workaround governance literature.
- Huntemann H, et al. Override rates of allergy alerts in commercial EP systems — representative published figures around 90%+ override rates.

**For the assessment**

The companion **Cases Module** on the live site gives you exam-grade practice. Use it after completing the self-test in Section 25. Apply the lecture frameworks — definitions, evidence appraisal, CDS design, governance — to two structured scenarios, one hospital and one community.

## SECTION 28 · CLOSING REFLECTION

# From Dr. Melhem

Electronic prescribing is the most powerful patient-safety intervention available to the modern hospital. It is also the most easily wasted. The hospitals that succeed are not those with the most expensive software — they are those that treat EP as a sociotechnical system: software shaped by workflows, workflows shaped by people, and people supported by culture and governance.

In Jordan we sit at an unusual moment. The infrastructure exists — Hakeem, the National ID, JFDA, our private hospital EHRs, our community pharmacy network. The connecting tissue does not yet. The pharmacy administrators graduating from this programme are the people who will build it.

Carry these five sentences from this lecture into your career: EP ≠ plug-and-play. Design determines benefit. Culture matters as much as code. Integrate, don't isolate. Monitor and adapt. And remember that across hospital and community, the pharmacist is the integrator — the only professional with both clinical knowledge and a daily stake in the medicines-data loop.

Thank you for your attention, your questions, and your engagement.

— **Dr. Samar J. Melhem, PhD**

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**End of study reference** · This document mirrors the 28-slide master lecture available at [samarjmelhem-maker.github.io/EP-Lecture-Dr-Samar-Melhem/](https://samarjmelhem-maker.github.io/EP-Lecture-Dr-Samar-Melhem/)